NAME

ovn-ctl − Open Virtual Network northbound daemon lifecycle utility

SYNOPSIS

ovn−ctl [*options*] *command* [−- *extra_args*]

DESCRIPTION

This program is intended to be invoked internally by Open Virtual Network startup scripts. System administrators should not normally invoke it directly.

COMMANDS

start_northd
start_controller
start_controller_vtep
start_ic
stop_northd
stop_controller
stop_controller_vtep
stop_ic
restart_northd
restart_controller
restart_controller_vtep
restart_ic
promote_ovnnb
promote_ovnsb
demote_ovnnb
demote_ovnsb
status_ovnnb
status_ovnsb
start_ovsdb
start_nb_ovsdb
start_sb_ovsdb
stop_ovsdb
stop_nb_ovsdb
stop_sb_ovsdb
restart_ovsdb
run_nb_ovsdb
run_sb_ovsdb
promote_ic_nb
promote_ic_sb
demote_ic_nb
demote_ic_sb
status_ic_nb
status_ic_sb
start_ic_ovsdb
start_ic_nb_ovsdb
start_ic_sb_ovsdb
stop_ic_ovsdb
stop_ic_nb_ovsdb
stop_ic_sb_ovsdb
restart_ic_ovsdb
run_ic_nb_ovsdb
run_ic_sb_ovsdb

start_ovnbr_ovsdb
start_br_controller
stop_ovnbr_ovsdb
stop_br_controller
restart_ovnbr_ovsdb
restart_br_controller
status_ovnbr_ovsdb
status_br_controller
run_ovnbr_ovsdb

OPTIONS

−−ovn−northd−priority=*NICE* sets the ovn-northd process niceness. The default is **−10**.

−−ovn−northd−wrapper=*WRAPPER* runs ovn-northd under the specified wrapper.

−−ovn−controller−priority=*NICE* sets the ovn-controller process niceness. The default is **−10**.

−−ovn−controller−wrapper=*WRAPPER* runs ovn-controller under the specified wrapper.

−−ovn−ic−priority=*NICE* sets the ovn-ic process niceness. The default is **−10**.

−−ovn−ic−wrapper=*WRAPPER* runs ovn-ic under the specified wrapper.

−−ovnbr−controller−priority=*NICE* sets the ovn-br-controller process niceness. The default is **−10**.

−−ovnbr−controller−wrapper=*WRAPPER* runs ovn-br-controller under the specified wrapper.

−−ovsdb−nb−wrapper=*WRAPPER* runs the Northbound database server under the specified wrapper.

−−ovsdb−sb−wrapper=*WRAPPER* runs the Southbound database server under the specified wrapper.

−−ovsdb−ovnbr−wrapper=*WRAPPER* runs the OVN bridge controller database server under the specified wrapper.

−−ovn−user=*USER[:GROUP]* runs OVN daemons as the specified user and optional group.

−−ovn−manage−ovsdb=*yes*|*no* controls whether **start_northd** and **stop_northd** also manage the NB and SB database servers. The default is **yes**.

−−ovn−controller−system−id=*SYSTEM_ID* sets the chassis name passed to ovn-controller. The default is empty.

−−ovsdb−disable−file−column−diff=*yes*|*no* controls whether database servers are started with **−−disable−file−column−diff**. The default is **no**.

−h | **−−help** displays usage information.

FILE LOCATION OPTIONS

−−db−sock=*SOCKET* sets the local Open vSwitch database socket.

−−db−nb−file=*FILE* sets the Northbound database file.

−−db−sb−file=*FILE* sets the Southbound database file.

−−db−nb−sock=*SOCKET* and **−−db−sb−sock=***SOCKET* set the NB and SB database sockets. Their defaults are **OVN_RUNDIR/ovnnb_db.sock** and **OVN_RUNDIR/ovnsb_db.sock**, respectively.

−−db−nb−schema=*FILE* sets the Northbound database schema file.

−−db−sb−schema=*FILE* sets the Southbound database schema file.

−−db−ic−nb−sock=*SOCKET* and **−−db−ic−sb−sock=***SOCKET* set the IC-NB and IC-SB database sockets. Their defaults are **OVN_RUNDIR/ovn_ic_nb_db.sock** and **OVN_RUNDIR/ovn_ic_sb_db.sock**, respectively.

The **−−db−nb−create−insecure−remote=***yes*|*no*, **−−db−sb−create−insecure−remote=***yes*|*no*, **−−db−ic−nb−create−insecure−remote=***yes*|*no*, **−−db−ic−sb−create−insecure−remote=***yes*|*no*, and

−−db−ovnbr−create−insecure−remote=*yes*|*no* options accept **yes** or **no**. The default is **no**. When set to **yes**, the database server creates an insecure ptcp remote; **no** does not create that remote.

−−db−ic−nb−file=*FILE* sets the IC-NB database file.

−−db−ic−sb−file=*FILE* sets the IC-SB database file.

−−db−ic−nb−schema=*FILE* sets the IC-NB database schema file.

−−db−ic−sb−schema=*FILE* sets the IC-SB database schema file.

−−db−nb−pidfile=*FILE* and **−−db−sb−pidfile=***FILE* set the NB and SB database pidfiles. Their defaults are **OVN_RUNDIR/ovnnb_db.pid** and **OVN_RUNDIR/ovnsb_db.pid**, respectively.

−−db−nb−ctrl−sock=*SOCKET* and **−−db−sb−ctrl−sock=***SOCKET* set the NB and SB database control sockets. Their defaults are **OVN_RUNDIR/ovnnb_db.ctl** and **OVN_RUNDIR/ovnsb_db.ctl**, respectively.

−−db−nb−config−file=*FILE* sets the Northbound database server configuration file.

−−db−sb−config−file=*FILE* sets the Southbound database server configuration file.

−−db−ic−nb−config−file=*FILE* sets the IC-NB database server configuration file.

−−db−ic−sb−config−file=*FILE* sets the IC-SB database server configuration file.

−−db−sb−relay−config−file=*FILE* sets the Southbound relay server configuration file.

−−db−nb−use−remote−in−db, **−−db−sb−use−remote−in−db**, **−−db−ic−nb−use−remote−in−db**, and **−−db−ic−sb−use−remote−in−db** accept **yes** or **no** and default to **yes**. A bare option is equivalent to specifying **yes**. The value **yes** makes ovsdb-server listen on remotes configured in the database **connections** column, while **no** does not. If a corresponding configuration file is specified, it takes precedence and this setting is ignored.

−−db−ovnbr−use−remote−in−db=*yes*|*no* controls whether ovsdb-server listens on the remotes configured in the database **connections** column. The default is **yes**.

−−ovn−controller−ssl−key=*KEY* sets the ovn-controller SSL/TLS private key file.

−−ovn−controller−ssl−cert=*CERT* sets the ovn-controller SSL/TLS certificate file.

−−ovn−controller−ssl−ca−cert=*CERT* sets the ovn-controller SSL/TLS CA certificate file.

−−ovn−controller−ssl−bootstrap−ca−cert=*CERT* bootstraps the ovn-controller SSL/TLS CA certificate file.

−−ovnbr−controller−ssl−key=*KEY* sets the ovn-br-controller SSL/TLS private key file.

−−ovnbr−controller−ssl−cert=*CERT* sets the ovn-br-controller SSL/TLS certificate file.

−−ovnbr−controller−ssl−ca−cert=*CERT* sets the ovn-br-controller SSL/TLS CA certificate file.

−−ovnbr−controller−ssl−bootstrap−ca−cert=*CERT* bootstraps the ovn-br-controller SSL/TLS CA certificate file.

−−ovn−ovnbr−db−ssl−key=*KEY* sets the ovn-br-controller database SSL/TLS private key file.

−−ovn−ovnbr−db−ssl−cert=*CERT* sets the ovn-br-controller database SSL/TLS certificate file.

−−ovn−ovnbr−db−ssl−ca−cert=*CERT* sets the ovn-br-controller database SSL/TLS CA certificate file.

−−db−ovnbr−sock=*SOCKET* sets the OVN bridge controller database socket.

−−db−ovnbr−file=*FILE* sets the OVN bridge controller database file.

−−db−ic−nb−pidfile=*FILE*, **−−db−ic−sb−pidfile=***FILE*, and **−−db−ovnbr−pidfile=***FILE* set the IC-NB, IC-SB, and OVN bridge controller database pidfiles. Their defaults are **OVN_RUNDIR/ovn_ic_nb_db.pid**, **OVN_RUNDIR/ovn_ic_sb_db.pid**, and **OVN_RUNDIR/ovnbr_db.pid**, respectively.

−−db−ic−nb−ctrl−sock=*SOCKET*, **−−db−ic−sb−ctrl−sock=***SOCKET*, and **−−db−ovnbr−ctrl−sock=***SOCKET* set database control sockets. Their defaults are **OVN_RUNDIR/ovn_ic_nb_db.ctl**, **OVN_RUNDIR/ovn_ic_sb_db.ctl**, and **OVN_RUNDIR/ovnbr_db.ctl**,

respectively.

−−db−ovnbr−schema=*FILE* sets the OVN bridge controller database schema file.

LOGGING OPTIONS

−−ovn−controller−log=*STRING*, **−−ovn−northd−log=***STRING*, and **−−ovn−ic−log=***STRING* set daemon logging parameters. The default is **−vconsole:emer −vsyslog:err −vfile:info**.

−−ovnbr−controller−log=*STRING* sets ovn-br-controller logging parameters. The default is **−vconsole:emer −vsyslog:err −vfile:info**.

−−ovn−nb−log=*STRING*, **−−ovn−sb−log=***STRING*, **−−ovn−ic−nb−log=***STRING*, **−−ovn−ic−sb−log=***STRING*, **−−ovn−sb−relay−log=***STRING*, and **−−ovn−ovnbr−log=***STRING* set database-server logging parameters. The defaults for the NB, SB, IC-NB, IC-SB, and OVN bridge controller database servers are **−vconsole:off −vfile:info**. The default for the SB relay database server is **−vconsole:emer −vsyslog:err −vfile:info**.

−−ovn−northd−logfile=*FILE*, **−−ovn−ic−logfile=***FILE*, **−−ovn−nb−logfile=***FILE*, **−−ovn−sb−logfile=***FILE*, **−−ovn−ic−nb−logfile=***FILE*, **−−ovn−ic−sb−logfile=***FILE*, **−−ovn−sb−relay−logfile=***FILE*, and **−−ovn−ovnbr−logfile=***FILE* select log files. Database-server log files default to files under **OVN_LOGDIR**; daemon log files default to empty.

−−ovn−northd−n−threads=*N_THREADS* sets the number of ovn-northd threads. The default is **1**.

PROTOCOL, CIPHER AND CIPHERSUITE OPTIONS

The daemon and database prefixes also accept their SSL/TLS key, certificate, and CA certificate options. The prefixes are **ovn−controller**, **ovn−northd**, **ovn−ic**, **ovn−nb−db**, **ovn−sb−db**, **ovn−ic−nb−db**, **ovn−ic−sb−db**, **ovn−sb−relay−db**, and **ovn−ovnbr−db**; append **−ssl−key**, **−ssl−cert**, or **−ssl−ca−cert**. Controller prefixes also accept **−ssl−bootstrap−ca−cert**. Empty values leave the normal database or daemon TLS settings in effect.

−−ovn−controller−ssl−protocols=*PROTOCOLS*.

−−ovn−ic−ssl−protocols=*PROTOCOLS*.

−−ovn−northd−ssl−protocols=*PROTOCOLS*.

−−ovn−nb−db−ssl−protocols=*PROTOCOLS*.

−−ovn−sb−db−ssl−protocols=*PROTOCOLS*.

−−ovn−ic−nb−db−ssl−protocols=*PROTOCOLS*.

−−ovn−ic−sb−db−ssl−protocols=*PROTOCOLS*.

−−ovn−controller−ssl−ciphers=*CIPHERS*.

−−ovn−ic−ssl−ciphers=*CIPHERS*.

−−ovn−northd−ssl−ciphers=*CIPHERS*.

−−ovn−nb−db−ssl−ciphers=*CIPHERS*.

−−ovn−sb−db−ssl−ciphers=*CIPHERS*.

−−ovn−ic−nb−db−ssl−ciphers=*CIPHERS*.

−−ovn−ic−sb−db−ssl−ciphers=*CIPHERS*.

−−ovn−controller−ssl−ciphersuites=*CIPHERSUITES*.

−−ovn−ic−ssl−ciphersuites=*CIPHERSUITES*.

−−ovn−northd−ssl−ciphersuites=*CIPHERSUITES*.

−−ovn−nb−db−ssl−ciphersuites=*CIPHERSUITES*.

−−ovn−sb−db−ssl−ciphersuites=*CIPHERSUITES*.

--ovn-ic-nb-db-ssl-ciphersuites=*CIPHERSUITES*.

--ovn-ic-sb-db-ssl-ciphersuites=*CIPHERSUITES*.

--ovn-controller-ssl-server-name=*NAME*.

--ovn-ic-ssl-server-name=*NAME*.

--ovn-northd-ssl-server-name=*NAME*.

--ovn-nb-db-ssl-server-name=*NAME*.

--ovn-sb-db-ssl-server-name=*NAME*.

--ovn-ic-nb-db-ssl-server-name=*NAME*.

--ovn-ic-sb-db-ssl-server-name=*NAME*.

--ovn-sb-relay-db-ssl-server-name=*NAME*.

--ovn-ovnbr-db-ssl-server-name=*NAME*.

--ovnbr-controller-ssl-protocols=*PROTOCOLS* sets the ovn-br-controller SSL/TLS protocols.

--ovnbr-controller-ssl-ciphers=*CIPHERS* sets the ovn-br-controller SSL/TLS cipher list.

--ovnbr-controller-ssl-ciphersuites=*CIPHERSUITES* sets the ovn-br-controller TLS 1.3 and later ciphersuite list.

--ovn-ovnbr-db-ssl-protocols=*PROTOCOLS* sets the ovn-br-controller database SSL/TLS protocols.

--ovn-ovnbr-db-ssl-ciphers=*CIPHERS* sets the ovn-br-controller database SSL/TLS cipher list.

--ovn-ovnbr-db-ssl-ciphersuites=*CIPHERSUITES* sets the ovn-br-controller database TLS 1.3 and later ciphersuite list.

ADDRESS AND PORT OPTIONS

--db-nb-sync-from-addr=*IP ADDRESS*.

--db-nb-sync-from-port=*PORT NUMBER*.

--db-nb-sync-from-proto=*PROTO*.

--db-sb-sync-from-addr=*IP ADDRESS*.

--db-sb-sync-from-port=*PORT NUMBER*.

--db-sb-sync-from-proto=*PROTO*.

--db-ic-nb-sync-from-addr=*IP ADDRESS*.

--db-ic-nb-sync-from-port=*PORT NUMBER*.

--db-ic-nb-sync-from-proto=*PROTO*.

--db-ic-sb-sync-from-addr=*IP ADDRESS*.

--db-ic-sb-sync-from-port=*PORT NUMBER*.

--db-ic-sb-sync-from-proto=*PROTO*.

The options **--db-nb-addr**, **--db-nb-port**, **--db-sb-addr**, **--db-sb-port**, **--db-ic-nb-addr**, **--db-ic-nb-port**, **--db-ic-sb-addr**, **--db-ic-sb-port**, **--db-ovnbr-addr**, and **--db-ovnbr-port** select the addresses and ports used when the corresponding **--db-*-create-insecure-remote** option is **yes**. When that option is **no**, these settings have no effect. The defaults for **--db-nb-addr** and **--db-nb-port** are **0.0.0.0** and **6641**, respectively. The defaults for **--db-sb-addr** and **--db-sb-port** are **0.0.0.0** and **6642**, respectively. The defaults for **--db-ic-nb-addr** and **--db-ic-nb-port** are **0.0.0.0** and **6645**, respectively. The defaults for **--db-ic-sb-addr** and **--db-ic-sb-port** are **0.0.0.0** and **6646**, respectively. The defaults for **--db-ovnbr-addr** and **--db-ovnbr-port** are **0.0.0.0** and **6651**, respectively.

--ovn-northd-nb-db and **--ovn-northd-sb-db** select the NB and SB remotes used by ovn-northd.

Similarly, **--ovn-ic-nb-db** and **--ovn-ic-sb-db** select the IC database remotes. Their defaults are the corresponding local unix database sockets.

--ovn-northd-nb-db=*PROTO*:*IP ADDRESS*: *PORT*..

--ovn-northd-sb-db=*PROTO*:*IP ADDRESS*: *PORT*..

--ovn-ic-nb-db=*PROTO*:*IP ADDRESS*: *PORT*..

--ovn-ic-sb-db=*PROTO*:*IP ADDRESS*: *PORT*..

CLUSTERING OPTIONS

--db-nb-cluster-local-addr=*IP ADDRESS*.

--db-nb-cluster-local-port=*PORT NUMBER*.

--db-nb-cluster-local-proto=*PROTO (tcp/ssl)*.

--db-nb-cluster-remote-addr=*IP ADDRESS*.

--db-nb-cluster-remote-port=*PORT NUMBER*.

--db-nb-cluster-remote-proto=*PROTO (tcp/ssl)*.

--db-nb-election-timer=*Timeout in milliseconds*.

--db-sb-cluster-local-addr=*IP ADDRESS*.

--db-sb-cluster-local-port=*PORT NUMBER*.

--db-sb-cluster-local-proto=*PROTO (tcp/ssl)*.

--db-sb-cluster-remote-addr=*IP ADDRESS*.

--db-sb-cluster-remote-port=*PORT NUMBER*.

--db-sb-cluster-remote-proto=*PROTO (tcp/ssl)*.

--db-sb-election-timer=*Timeout in milliseconds*.

--db-ic-nb-cluster-local-addr=*IP ADDRESS*.

--db-ic-nb-cluster-local-port=*PORT NUMBER*.

--db-ic-nb-cluster-local-proto=*PROTO (tcp/ssl)*.

--db-ic-nb-cluster-remote-addr=*IP ADDRESS*.

--db-ic-nb-cluster-remote-port=*PORT NUMBER*.

--db-ic-nb-cluster-remote-proto=*PROTO (tcp/ssl)*.

--db-ic-sb-cluster-local-addr=*IP ADDRESS*.

--db-ic-sb-cluster-local-port=*PORT NUMBER*.

--db-ic-sb-cluster-local-proto=*PROTO (tcp/ssl)*.

--db-ic-sb-cluster-remote-addr=*IP ADDRESS*.

--db-ic-sb-cluster-remote-port=*PORT NUMBER*.

--db-ic-sb-cluster-remote-proto=*PROTO (tcp/ssl)*.

--db-cluster-schema-upgrade=*yes*|*no* controls whether clustered database servers automatically upgrade their schemas when they start. The default is **yes**; specify **no** to disable the automatic upgrade.

--no-db-cluster-schema-upgrade is equivalent to **--db-cluster-schema-upgrade=no**. The default is **yes**.

PROBE INTERVAL OPTIONS

--db-nb-probe-interval-to-active=*Time in milliseconds* .

−−db−sb−probe−interval−to−active=*Time in milliseconds* .

RELAY OPTIONS

−−db−sb−relay−sock=*SOCKET*, **−−db−sb−relay−pidfile=***FILE*, and **−−db−sb−relay−ctrl−sock=***SOCKET* set the relay socket, pidfile, and control socket. The defaults are **ovnsb_relay_db.sock**, **ovnsb_relay_db.pid**, and **ovnsb_relay_db.ctl** in **OVN_RUNDIR**.

−−db−sb−relay−remote=*REMOTE* selects the upstream relay remote. The default is empty.

−−db−sb−relay−use−remote−in−db=*yes*|*no* controls whether ovsdb-server listens on the Southbound remotes configured in the database **connections** column. The default is **yes**. A bare option is equivalent to specifying **yes**.

EXTRA OPTIONS

Any options after '−' will be passed on to the binary run by *command* with the exception of start_northd, which can have options specified in ovn-northd-db-params.conf. Any *extra_args* passed to start_northd will be passed to the ovsdb-servers if **−−ovn−manage−ovsdb=yes**

CONFIGURATION FILES

Following are the optional configuration files. If present, it should be located in the etc dir

ovnnb−active.conf

If present, this file should hold the url to connect to the active Northbound DB server

tcp:x.x.x.x:6641

ovnsb−active.conf

If present, this file should hold the url to connect to the active Southbound DB server

tcp:x.x.x.x:6642

ovn−northd−db−params.conf

If present, start_northd will not start the DB server even if **−−ovn−manage−ovsdb=yes**. This file should hold the database url parameters to be passed to ovn-northd.

−−ovnnb−db=tcp:x.x.x.x:6641 −−ovnsb−db=tcp:x.x.x.x:6642

ic−nb−active.conf

If present, this file should hold the url to connect to the active Interconnection Northbound DB server

tcp:x.x.x.x:6645

ic−sb−active.conf

If present, this file should hold the url to connect to the active Interconnection Southbound DB server

tcp:x.x.x.x:6646

ovn−ic−db−params.conf

If present, this file should hold the database url parameters to be passed to ovn-ic.

−−ic−nb−db=tcp:x.x.x.x:6645 −−ic−sb−db=tcp:x.x.x.x:6646

RUNNING OVN DB SERVERS WITHOUT DETACHING

 # ovn−ctl run_nb_ovsdb

This command runs the OVN nb ovsdb-server without passing the **detach** option, making it to block until ovsdb-server exits. This command will be useful for starting the OVN nb ovsdb-server in a container.

 # ovn−ctl run_sb_ovsdb

This command runs the OVN sb ovsdb-server without passing the **detach** option, making it to block until ovsdb-server exits. This command will be useful for starting the OVN sb ovsdb-server in a container.

 # ovn−ctl run_ic_nb_ovsdb

This command runs the OVN IC-NB ovsdb-server without passing the **detach** option, making it to block until ovsdb-server exits. This command will be useful for starting the OVN IC-NB ovsdb-server in a container.

 # ovn−ctl run_ic_sb_ovsdb

This command runs the OVN IC-SB ovsdb-server without passing the **detach** option, making it to block until ovsdb-server exits. This command will be useful for starting the OVN IC-SB ovsdb-server in a container.

 # ovn−ctl run_ovnbr_ovsdb

This command runs the OVN bridge db ovsdb-server without passing the **detach** option, making it to block until ovsdb-server exits. This command will be useful for starting the OVN br db ovsdb-server in a container.

EXAMPLE USAGE

Run ovn−controller on a host already running OVS
 # ovn−ctl start_controller

Run ovn−northd on a host already running OVS
 # ovn−ctl start_northd

All−in−one OVS+OVN for testing
 # ovs−ctl start −−system−id="random"

 # ovn−ctl start_northd

 # ovn−ctl start_controller

Promote and demote ovsdb servers
 # ovn−ctl promote_ovnnb

 # ovn−ctl promote_ovnsb

 # ovn−ctl −−db−nb−sync−from−addr=x.x.x.x −−db−nb−sync−from−port=6641 −−db−nb−probe−interval−to−active=60000 demote_ovnnb

 # ovn−ctl −−db−sb−sync−from−addr=x.x.x.x −−db−sb−sync−from−port=6642 −−db−sb−probe−interval−to−active=60000 demote_ovnsb

Creating a clustered db on 3 nodes with IPs x.x.x.x, y.y.y.y and z.z.z.z
Starting OVN ovsdb servers and ovn-northd on the node with IP x.x.x.x

 # ovn−ctl −−db−nb−addr=x.x.x.x −−db−nb−create−insecure−remote=yes −−db−sb−addr=x.x.x.x −−db−sb−create−insecure−remote=yes −−db−nb−cluster−local−addr=x.x.x.x −−db−sb−cluster−local−addr=x.x.x.x −−ovn−northd−nb−db=tcp:x.x.x.x:6641,tcp:y.y.y.y:6641,tcp:z.z.z.z:6641 −−ovn−northd−sb−db=tcp:x.x.x.x:6642,tcp:y.y.y.y:6642,tcp:z.z.z.z:6642 start_northd

Starting OVN ovsdb-servers and ovn-northd on the node with IP y.y.y.y and joining the cluster started at x.x.x.x

 # ovn−ctl −−db−nb−addr=y.y.y.y −−db−nb−create−insecure−remote=yes −−db−sb−addr=y.y.y.y −−db−sb−create−insecure−remote=yes −−db−nb−cluster−local−addr=y.y.y.y −−db−sb−cluster−local−addr=y.y.y.y −−db−nb−cluster−remote−addr=x.x.x.x −−db−sb−cluster−remote−addr=x.x.x.x −−ovn−northd−nb−db=tcp:x.x.x.x:6641,tcp:y.y.y.y:6641,tcp:z.z.z.z:6641 −−ovn−northd−sb−db=tcp:x.x.x.x:6642,tcp:y.y.y.y:6642,tcp:z.z.z.z:6642 start_northd

Starting OVN ovsdb-servers and ovn-northd on the node with IP z.z.z.z and joining the cluster started at x.x.x.x

 # ovn−ctl −−db−nb−addr=z.z.z.z −−db−nb−create−insecure−remote=yes −−db−nb−cluster−local−addr=z.z.z.z −−db−sb−addr=z.z.z.z −−db−sb−create−insecure−remote=yes −−db−sb−cluster−local−addr=z.z.z.z −−db−nb−cluster−remote−addr=x.x.x.x −−db−sb−cluster−remote−addr=x.x.x.x −−ovn−northd−nb−db=tcp:x.x.x.x:6641,tcp:y.y.y.y:6641,tcp:z.z.z.z:6641 −−ovn−northd−sb−db=tcp:x.x.x.x:6642,tcp:y.y.y.y:6642,tcp:z.z.z.z:6642 start_northd

Passing ssl keys when starting OVN dbs will supersede the default ssl values in db
Starting standalone ovn db server passing SSL/TLS certificates

 # ovn−ctl −−ovn−nb−db−ssl−key=/etc/ovn/ovnnb−privkey.pem −−ovn−nb−db−ssl−cert=/etc/ovn/ovnnb−cert.pem −−ovn−nb−db−ssl−ca−cert=/etc/ovn/cacert.pem −−ovn−sb−db−ssl−key=/etc/ovn/ovnsb−privkey.pem −−ovn−sb−db−ssl−cert=/etc/ovn/ovnsb−cert.pem

−−ovn−sb−db−ssl−ca−cert=/etc/ovn/cacert.pem start_northd

Avoiding automatic clustered OVN database schema upgrade

If you desire more control over clustered DB schema upgrade, you can opt-out of automatic on-start upgrade attempts with **−−no−db−cluster−schema−upgrade**.

Start OVN NB and SB clustered databases on host with IP x.x.x.x without schema upgrade

 # ovn−ctl start_nb_ovsdb −−db−nb−cluster−local−addr=x.x.x.x −−no−db−cluster−schema−upgrade # ovn−ctl start_sb_ovsdb −−db−sb−cluster−local−addr=x.x.x.x −−no−db−cluster−schema−upgrade

Trigger clustered DB schema upgrade manually

 # ovsdb−client convert unix:/var/run/ovn/ovnnb_db.sock /usr/local/share/ovn/ovn−nb.ovsschema # ovsdb−client convert unix:/var/run/ovn/ovnsb_db.sock /usr/local/share/ovn/ovn−sb.ovsschema

Run OVN bridge controller services on a host already running OVS

 # ovn−ctl start_ovnbr_ovsdb

 # ovn−ctl start_br_controller